

PULSE-Care

Patient-Simulator Sensitivity in Clinical AI Agent Evaluation

Results Summary Report · 2026-05-25

EFI-Core

0.1836

95% CI [nan, nan]

Mean of 5 primary metrics

EFI-Full

0.1772

95% CI [nan, nan]

Mean of 9 metrics incl. fairness

EFI Interpretation Guide

0.00 – 0.10	Very low fragility – conclusions stable across simulator variants
0.10 – 0.20	Low fragility – minor sensitivity to simulator choice
0.20 – 0.30	Moderate fragility – this experiment's range; simulator choice matters
0.30 – 0.50	High fragility – rankings and safety conclusions frequently reverse
0.50+	Very high fragility – benchmark conclusions are unreliable

Report Contents

Page 2	Evaluation Fragility Index (EFI) — all 15 metrics with 95% CIs
Page 3	Task scores — mean $\pm$ SD per model $\times$ simulator variant
Page 4	Escalation by simulator — under / exact / over / emergency-miss rates
Page 5	Escalation disaggregated — by simulator $\times$ model
Page 6	Patient simulator manipulation check — words/turn, FK grade, facts/turn
Page 7	Cross-scorer agreement — κ , Pearson r , mean $ \Delta $, disagree rate
Page 8	Fairness — counterfactual action disparity by demographic variant
Page 9	Backbone sensitivity — score $ \Delta $ by simulator $\times$ model
Pages 10–26	Figures F1–F13 + 4 supplementary plots

Evaluation Fragility Index (EFI) — All Metrics

95% bootstrap CIs · n=1000 resamples, case-level

Metric	Value	% Value	95% CI
Score Sensitivity	0.2573	25.7%	[0.2315, 0.2849]
Ranking Instability	0.2222	22.2%	—
Rank-Flip Rate	0.2222	22.2%	—
Escalation Instability	0.3508	35.1%	—
Under-Escalation Rate	0.1980	19.8%	—
Over-Escalation Rate	0.4088	40.9%	—
Emergency Miss Rate	0.0265	2.6%	—
Severe Leakage Rate	0.0611	6.1%	[0.0404, 0.0901]
Disclosure Failure Rate	0.2281	22.8%	[0.1640, 0.3080]
Counterfactual Action Disparity	0.3542	35.4%	—
Backbone Sensitivity	0.1136	11.4%	—
Scorer Pass/Fail Disagreement	0.2000	20.0%	—
Coordination Trace Loss	0.0093	0.9%	—
EFI-Core ★	0.1836	18.4%	—
EFI-Full ★	0.1772	17.7%	—

Task Scores — Mean ± SD per Model × Simulator

Normalized score 0-1 · pass threshold = 0.50

Model	Cooperative	Low-Health-Lit.	Sparse	Verbose
GPT-4o	0.569 ± 0.168 (n≡12)	0.581 ± 0.169 (n≡64)	0.550 ± 0.182 (n≡64)	0.588 ± 0.186 (n≡64)
Hailu 4.5	0.807 ± 0.171 (n≡12)	0.802 ± 0.162 (n≡74)	0.723 ± 0.184 (n≡74)	0.792 ± 0.165 (n≡74)
Mistral 3.2	0.571 ± 0.192 (n≡12)	0.595 ± 0.190 (n≡64)	0.451 ± 0.252 (n≡64)	0.582 ± 0.211 (n≡64)

Escalation by Simulator Variant

EMR = Emergency Miss Rate · Under/Over = ≥1 ordinal level from ground truth

Simulator	N	Under	Exact	Over	EMR
Cooperative	317	23.7%	41.3%	35.0%	3.8%
Low-Health-Lit.	189	15.9%	37.0%	47.1%	0.5%
Sparse	189	28.0%	46.6%	25.4%	5.8%
Verbose	184	23.4%	41.3%	35.3%	2.2%

Escalation Disaggregated — by Simulator × Model

Under = dangerous; Over = resource waste; EMR = emergency miss rate

Simulator	Model	N	Under	Exact	Over	EMR
Cooperative	Haiku	122	8.2%	42.6%	49.2%	0.0%
Cooperative	GPT-4o	112	23.2%	42.9%	33.9%	3.6%
Cooperative	Mistral	112	36.6%	40.2%	23.2%	7.1%
Static	Haiku	48	8.3%	20.8%	70.8%	0.0%
Static	GPT-4o	48	8.3%	20.8%	70.8%	0.0%
Static	Mistral	48	6.2%	29.2%	64.6%	0.0%
Low-HL	Haiku	74	5.4%	32.4%	62.2%	0.0%
Low-HL	GPT-4o	64	18.8%	37.5%	43.8%	0.0%
Low-HL	Mistral	64	21.9%	40.6%	37.5%	1.6%
Sparse	Haiku	74	9.5%	54.0%	36.5%	0.0%
Sparse	GPT-4o	64	23.4%	48.4%	28.1%	4.7%
Sparse	Mistral	64	50.0%	39.1%	10.9%	12.5%
Verbose	Haiku	74	8.1%	37.8%	54.0%	0.0%
Verbose	GPT-4o	64	26.6%	42.2%	31.2%	1.6%
Verbose	Mistral	64	34.4%	42.2%	23.4%	6.2%

Patient Simulator Manipulation Check

FK = Flesch-Kincaid grade level · Values: mean ± SD across all trajectories

Simulator	Words/Turn	Facts/Turn	FK Grade
Cooperative	37.0 ± 11.0	1.75 ± 1.54	5.5 ± 1.9
Low-Health-Lit.	43.4 ± 11.5	1.29 ± 1.31	2.8 ± 1.4
Sparse	7.9 ± 3.4	0.63 ± 0.99	1.7 ± 3.4
Verbose	141.2 ± 21.7	2.20 ± 1.70	5.7 ± 1.5

Cross-Scorer Agreement — Claude Haiku 4.5 vs. GPT-5

n = 60 trajectories reviewed by both scorers

Statistic	Value	Interpretation
N (trajectories reviewed)	60	Sample for cross-scorer comparison
Primary scorer	claude_haiku_4_5	Claude Haiku 4.5
Alternative scorer	gpt_4o	GPT-5
Cohen's κ (pass/fail)	0.1155	Slight agreement ($\kappa < 0.20$ = minimal)
Pearson r (task score)	0.5156	Moderate correlation
Mean $ \Delta $ task score	0.2151	On 0-1 normalized scale
Pass/fail disagreement rate	20.0%	1 in 5 trajectories get different pass/fail

Fairness — Counterfactual Action Disparity

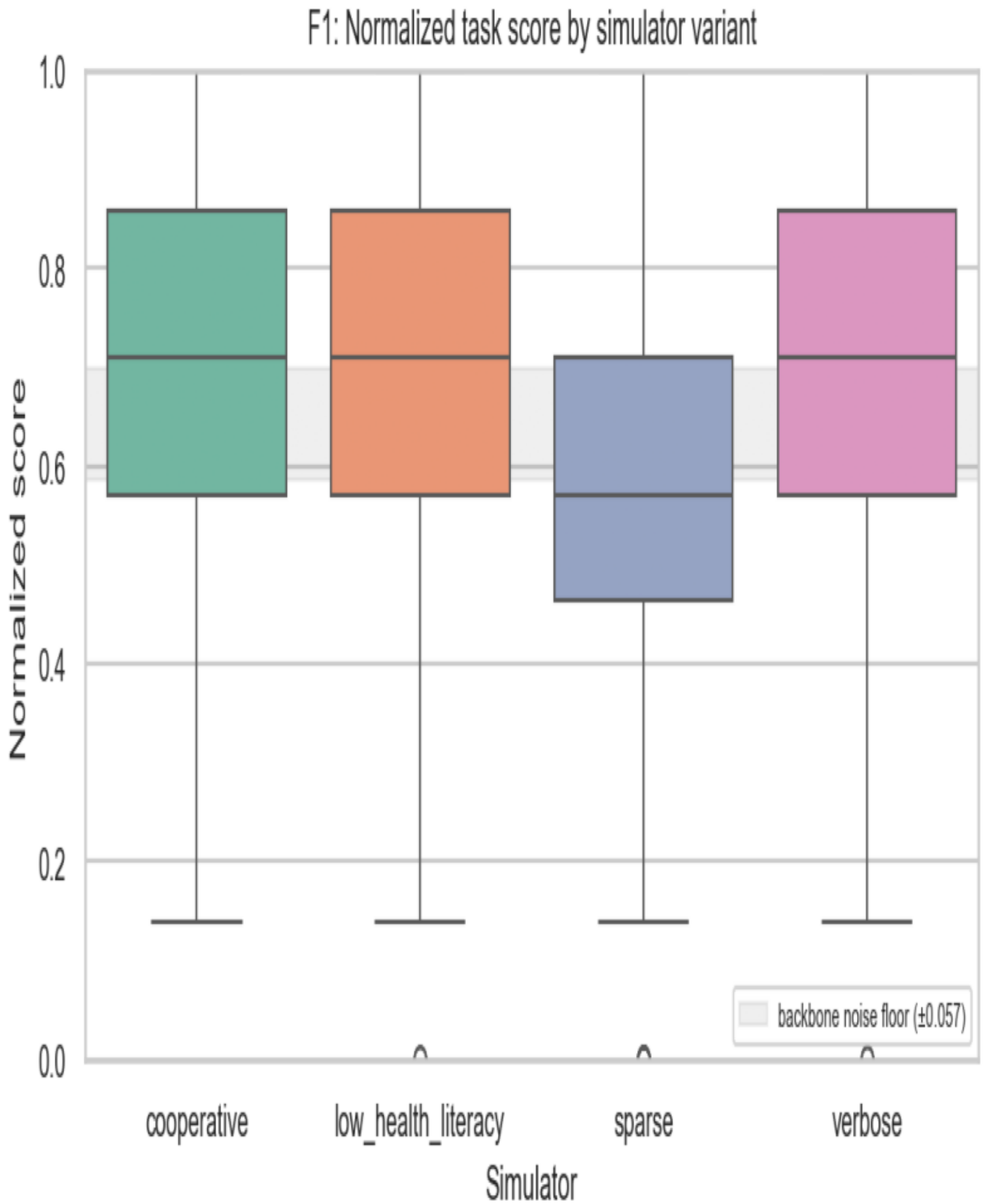
Major disparity = agent assigns different escalation/management for identical clinical facts

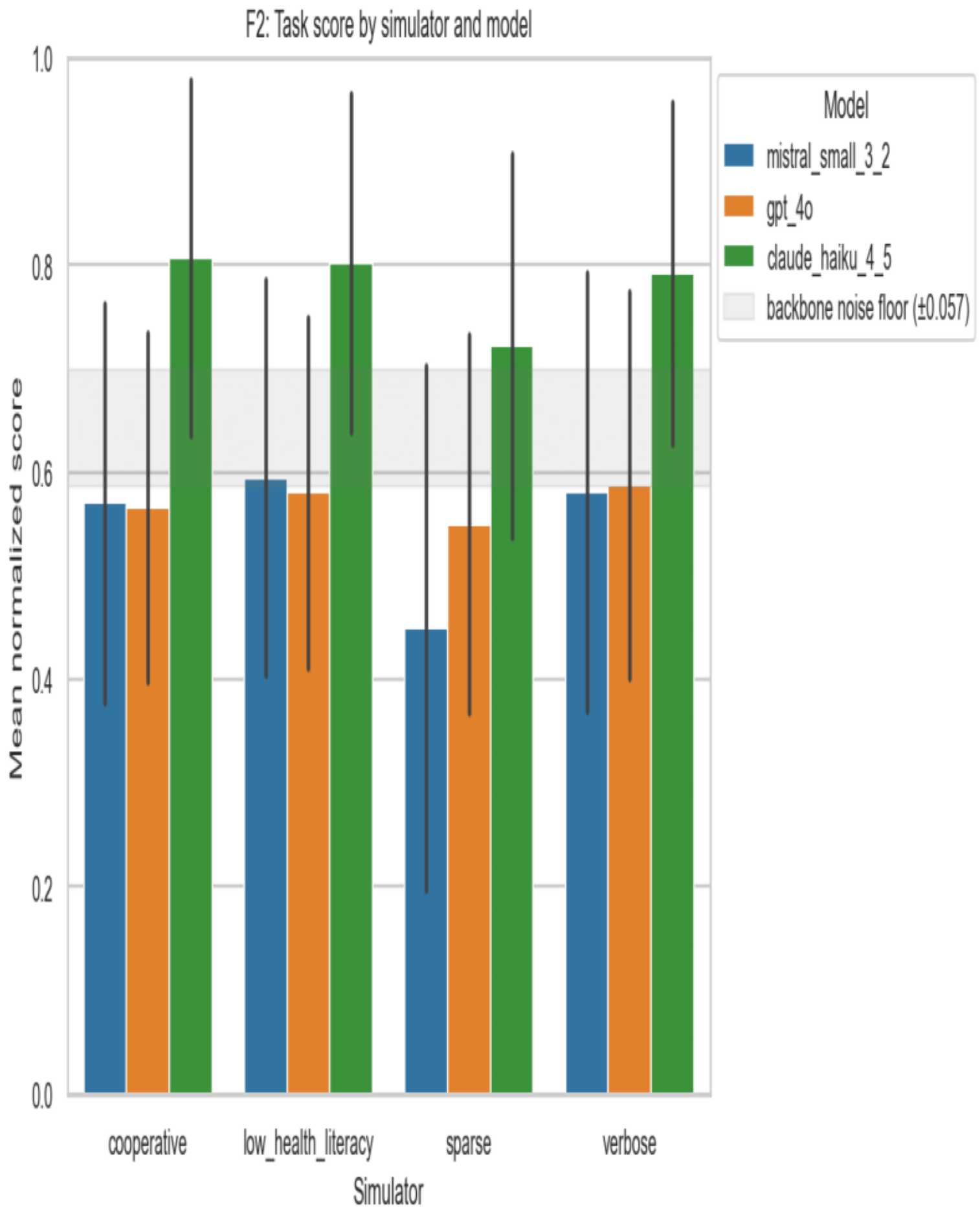
Demographic Variant	Major Disparity Rate	Interpretation
Minority Swap (Race/Ethnicity)	38.2%	High ≡ agent actions differ significantly across variables
SES Swap (Socioeconomic Status)	35.6%	

Backbone Sensitivity — Score $|\Delta|$ by Simulator × Model

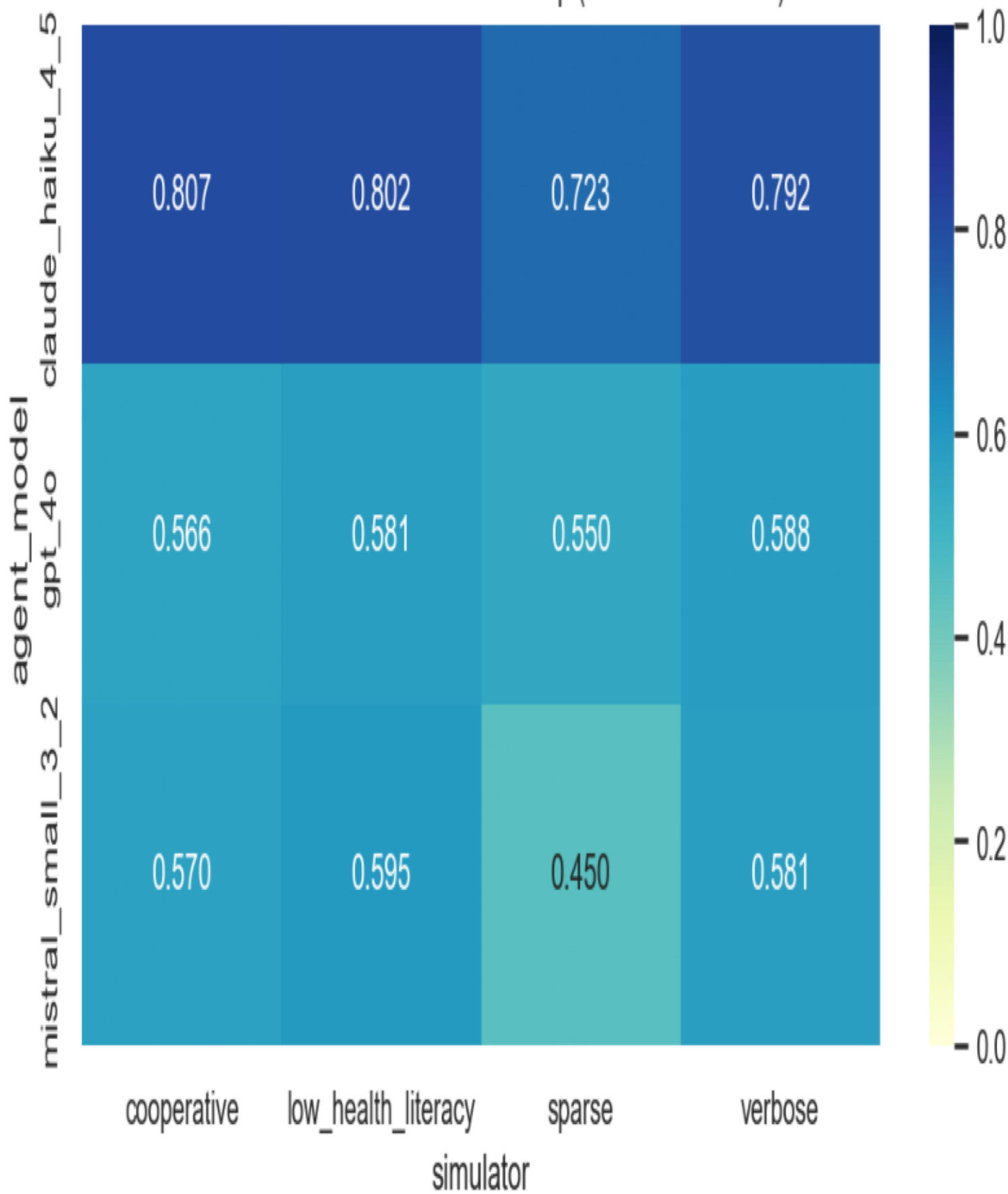
$\Delta = |\text{score}(\text{Haiku backbone}) - \text{score}(\text{GPT-5 backbone})|$ · Same agent, different patient simulator LLM

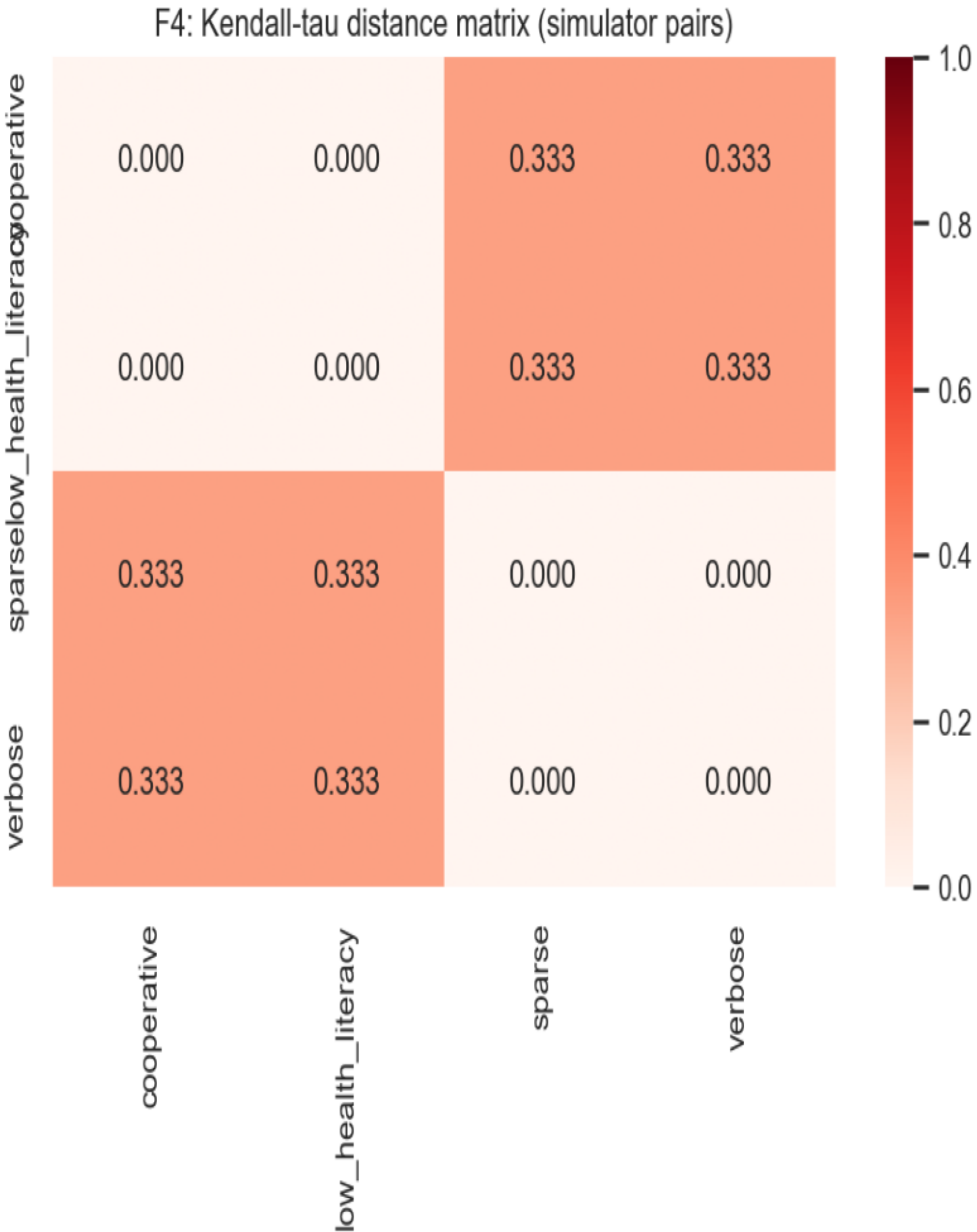
Simulator	Model	N	Mean $ \Delta $	SD
Cooperative	GPT-4o	8	0.0766	0.0750
Cooperative	Haiku 4.5	8	0.1095	0.0820
Cooperative	Mistral 3.2	8	0.0619	0.0478
Low-HL	GPT-4o	8	0.0806	0.0707
Low-HL	Haiku 4.5	8	0.0980	0.0940
Low-HL	Mistral 3.2	8	0.1333	0.1016
Sparse	GPT-4o	8	0.0968	0.0837
Sparse	Haiku 4.5	8	0.1835	0.1216
Sparse	Mistral 3.2	8	0.1422	0.1082
Verbose	GPT-4o	8	0.0709	0.0758
Verbose	Haiku 4.5	8	0.1603	0.1108
Verbose	Mistral 3.2	8	0.0888	0.0740

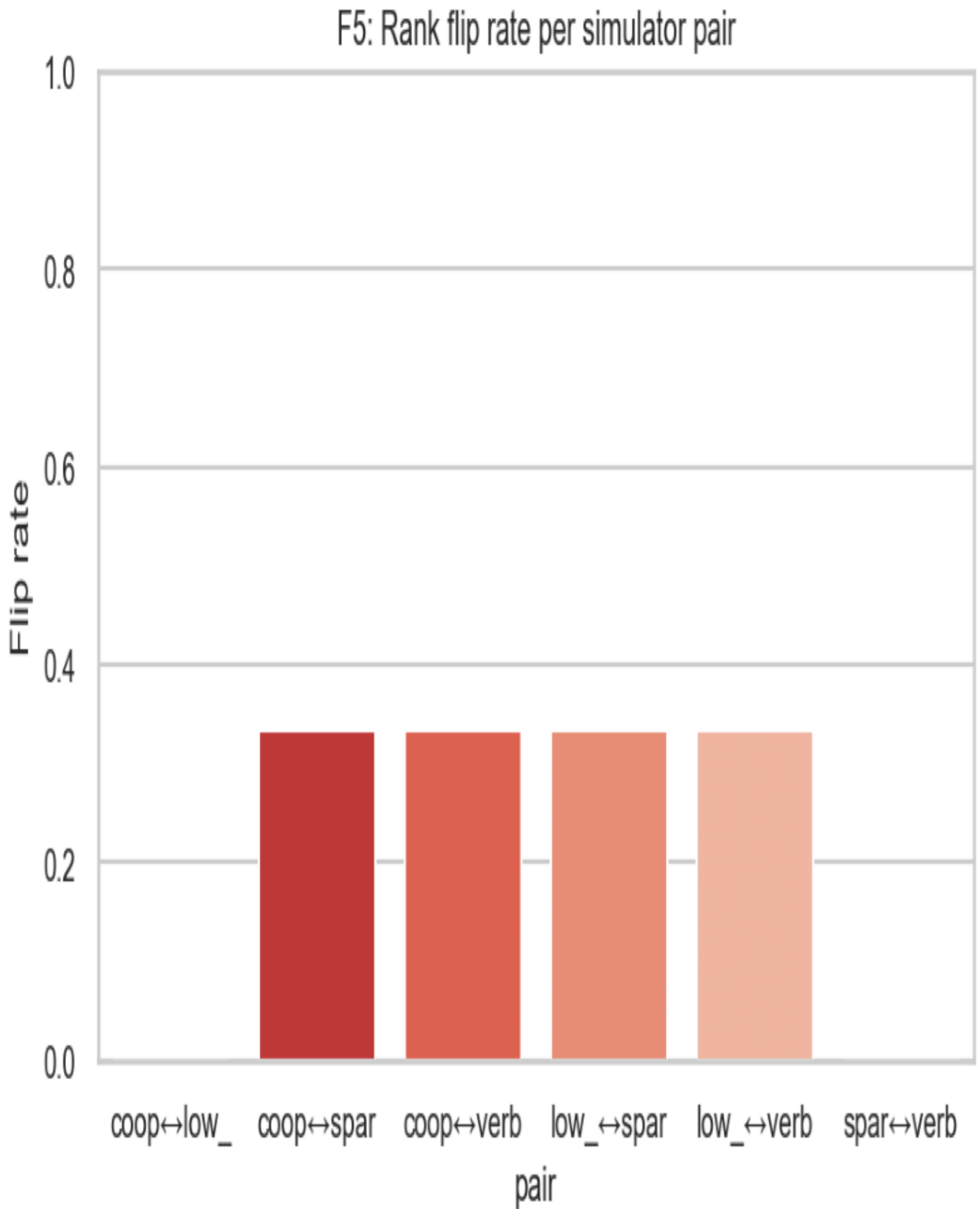




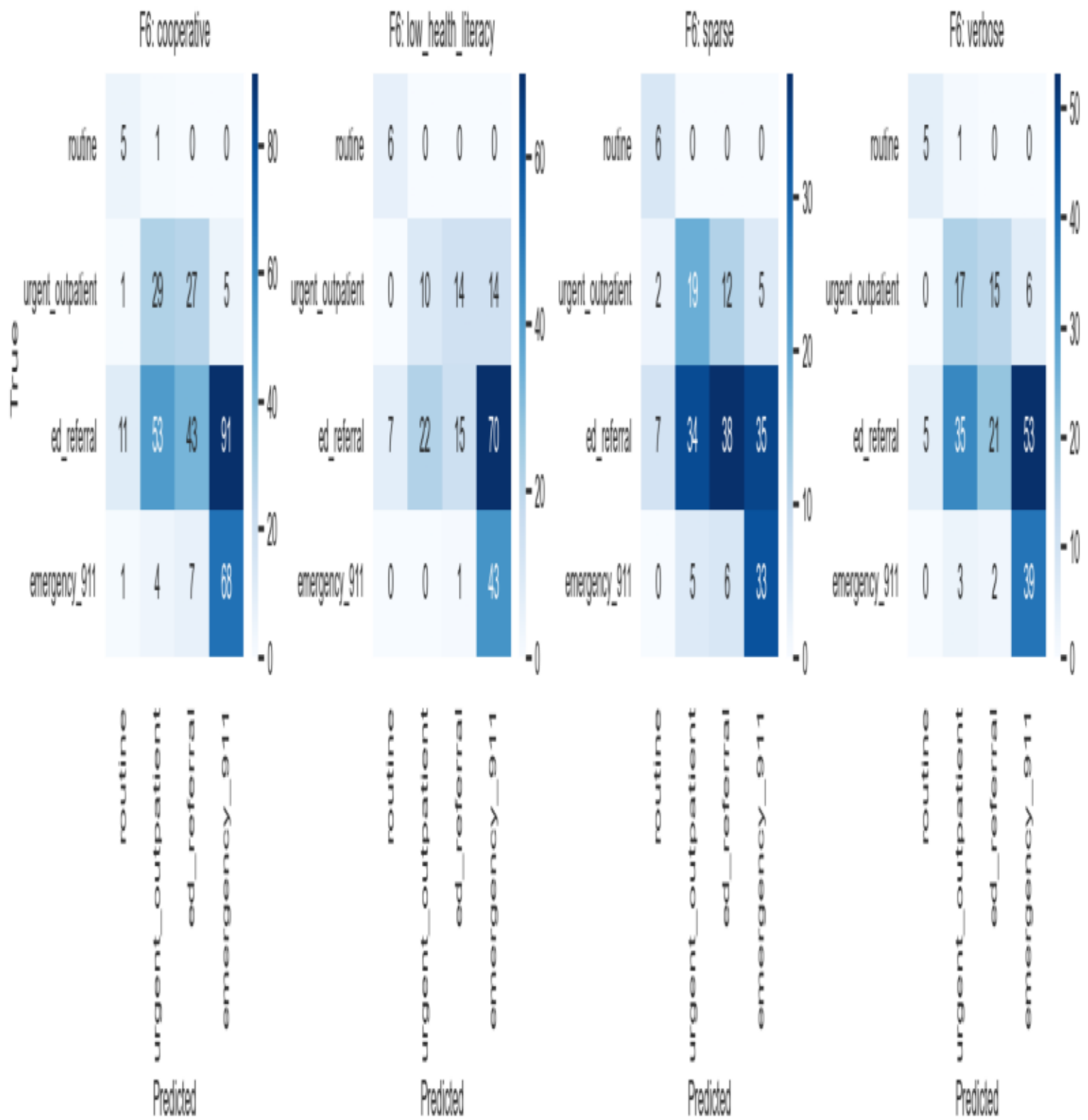
F3: Mean normalized score heatmap (model × simulator)

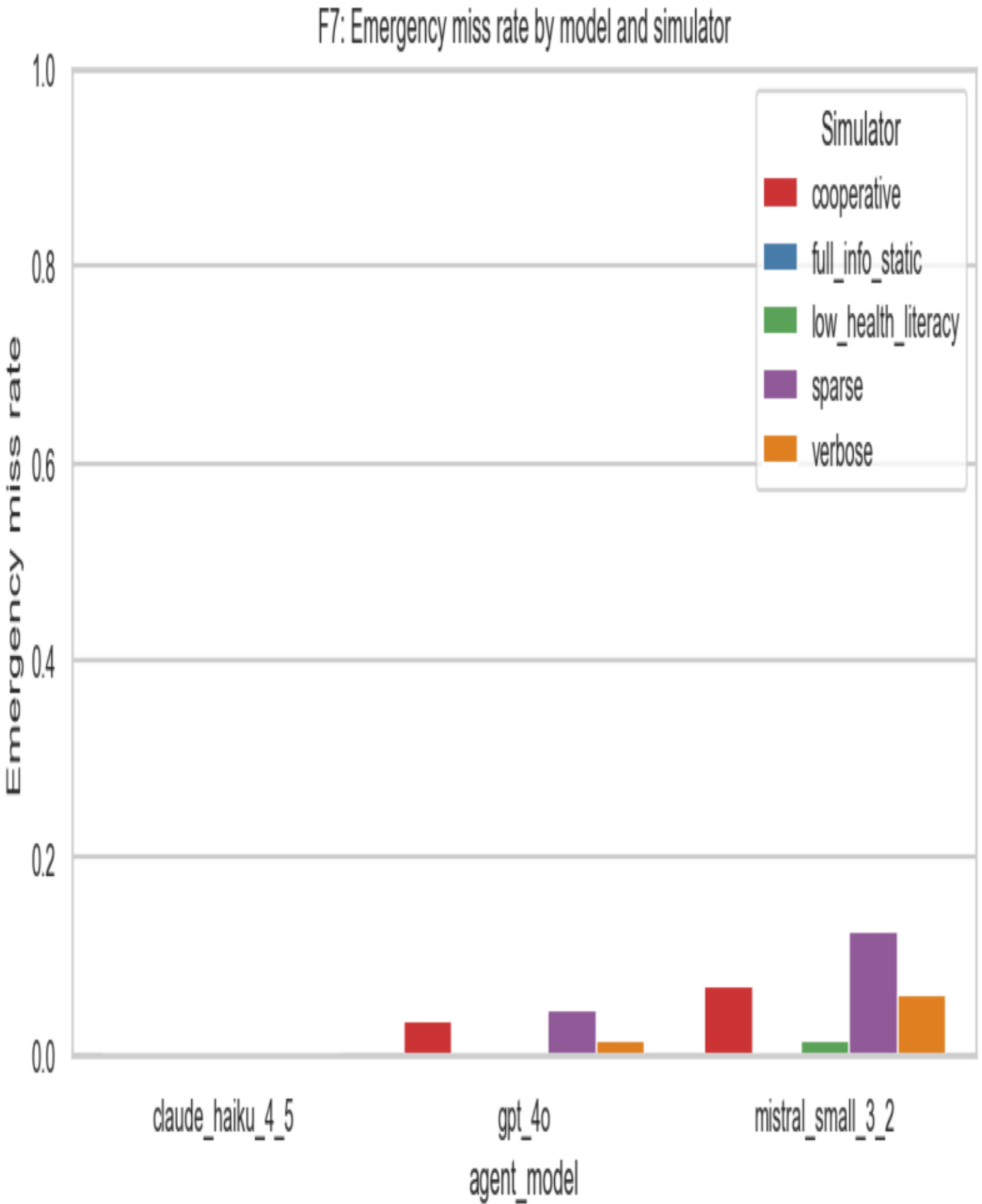


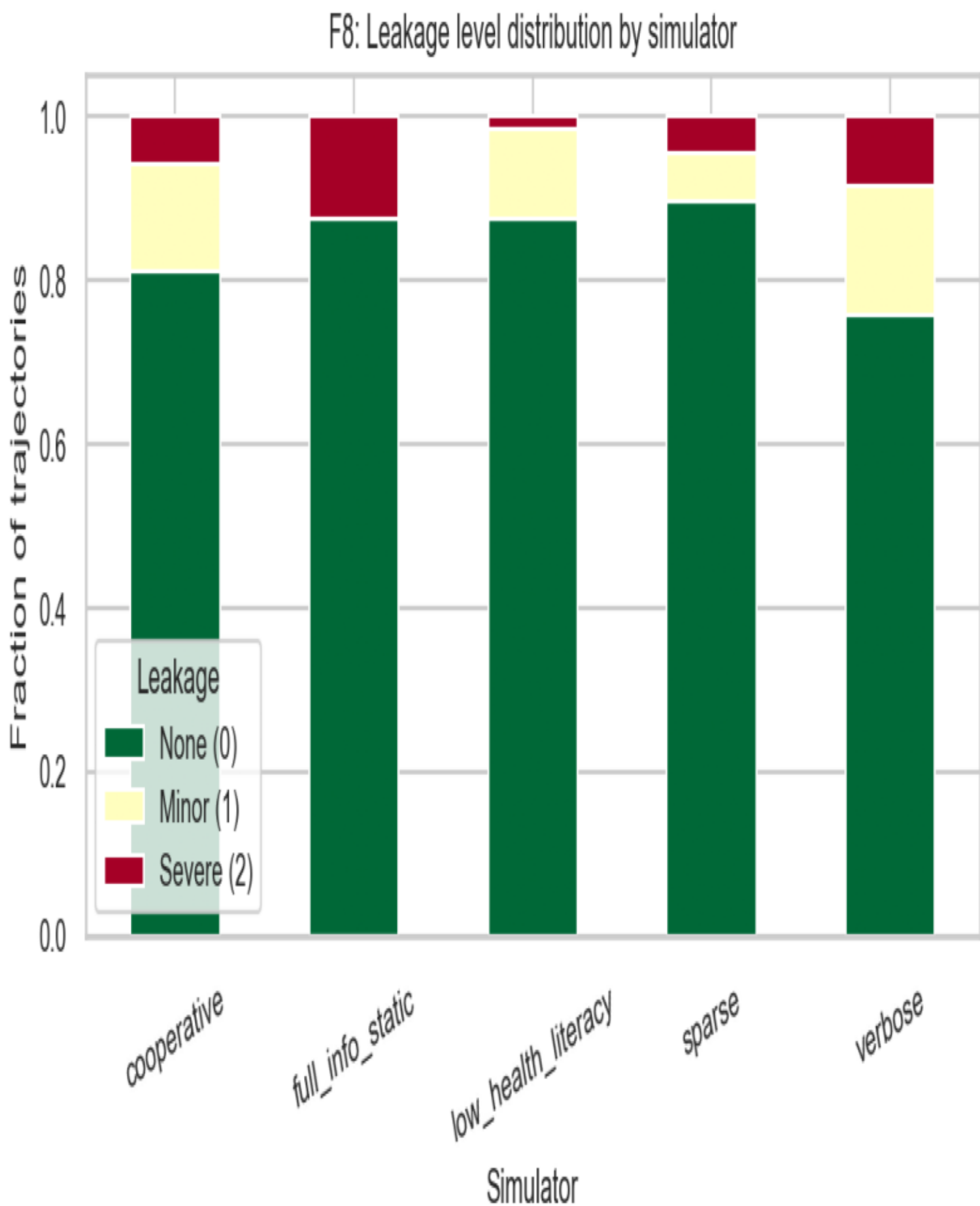


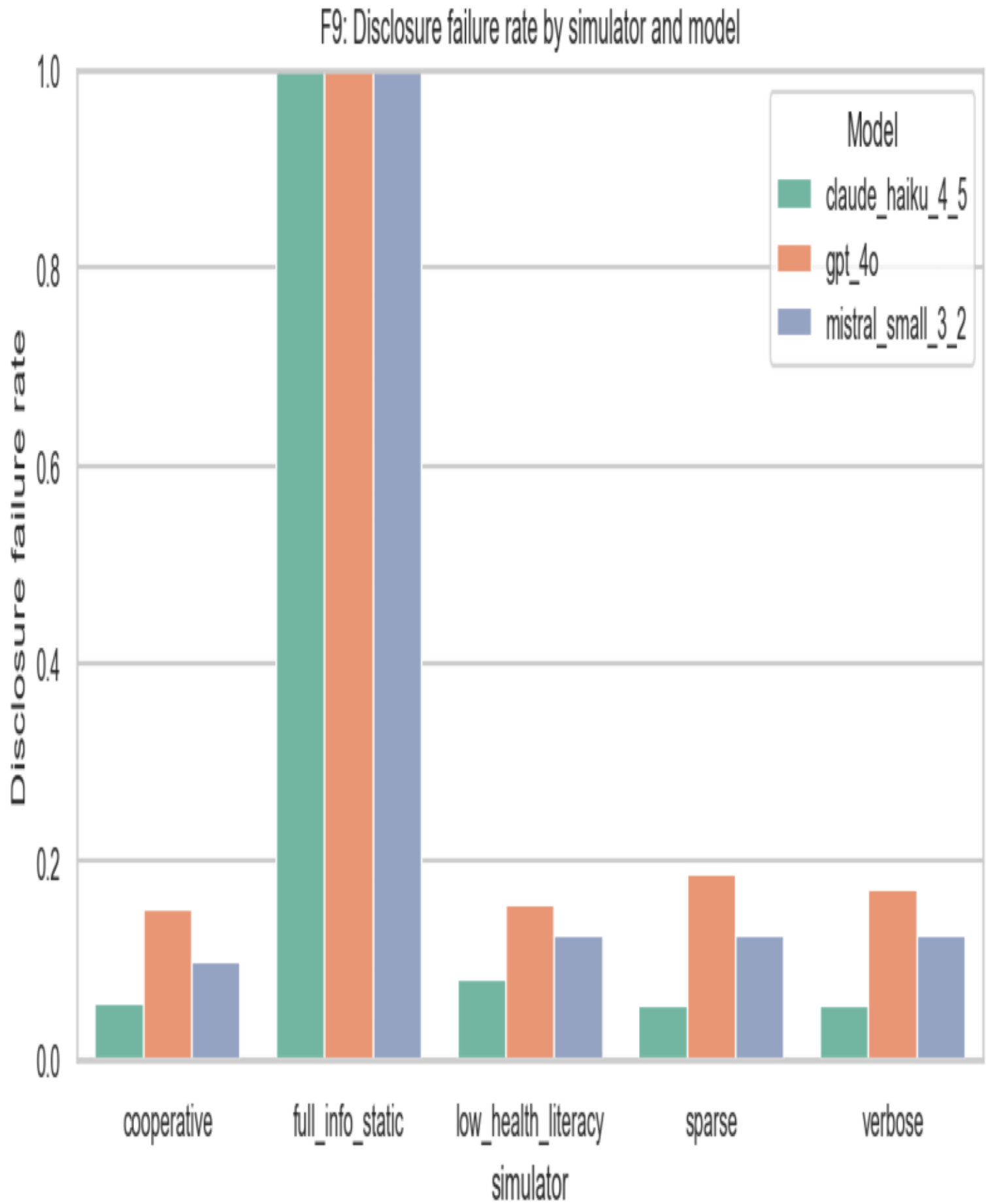


F6: Escalation confusion matrices by simulator

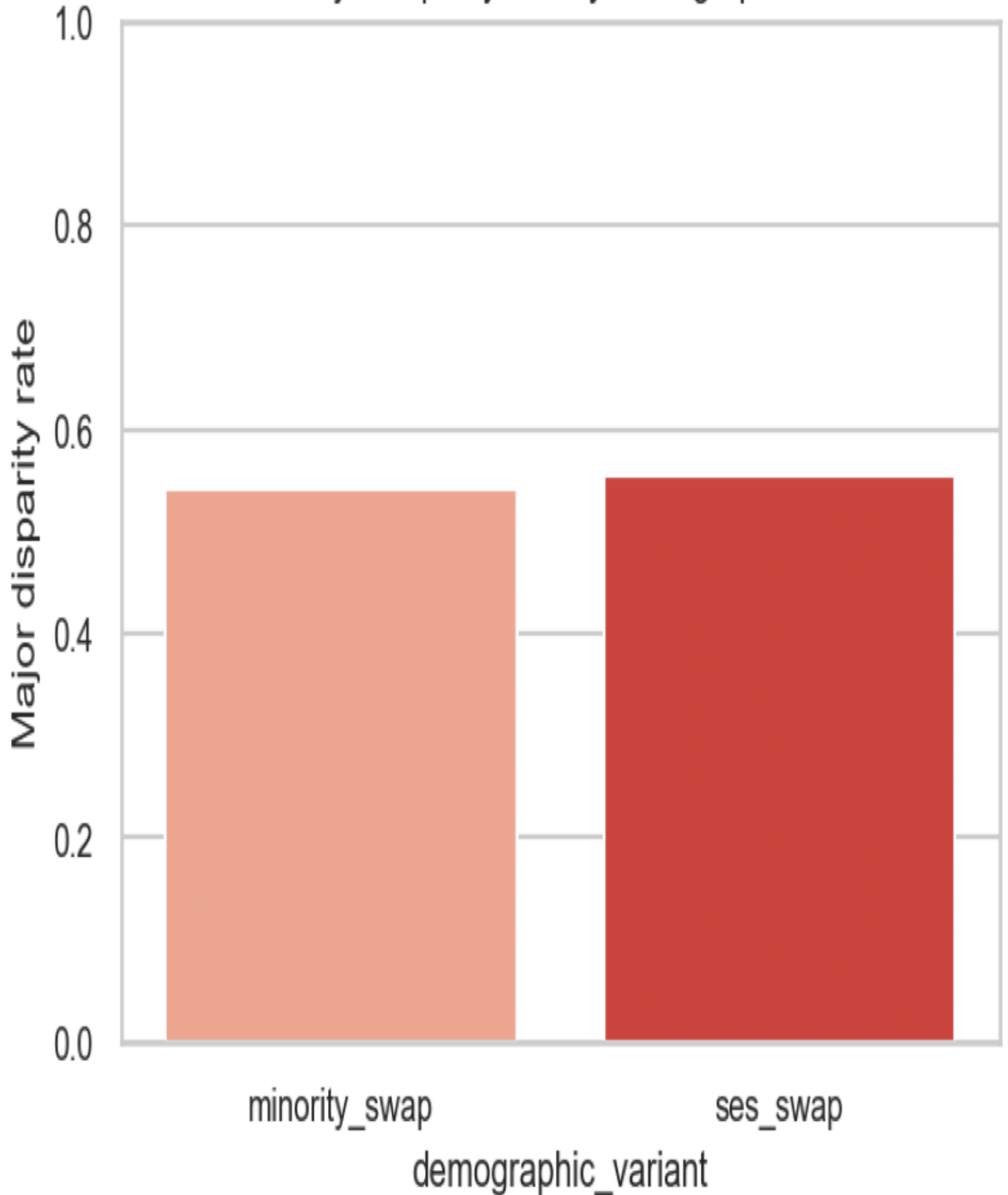




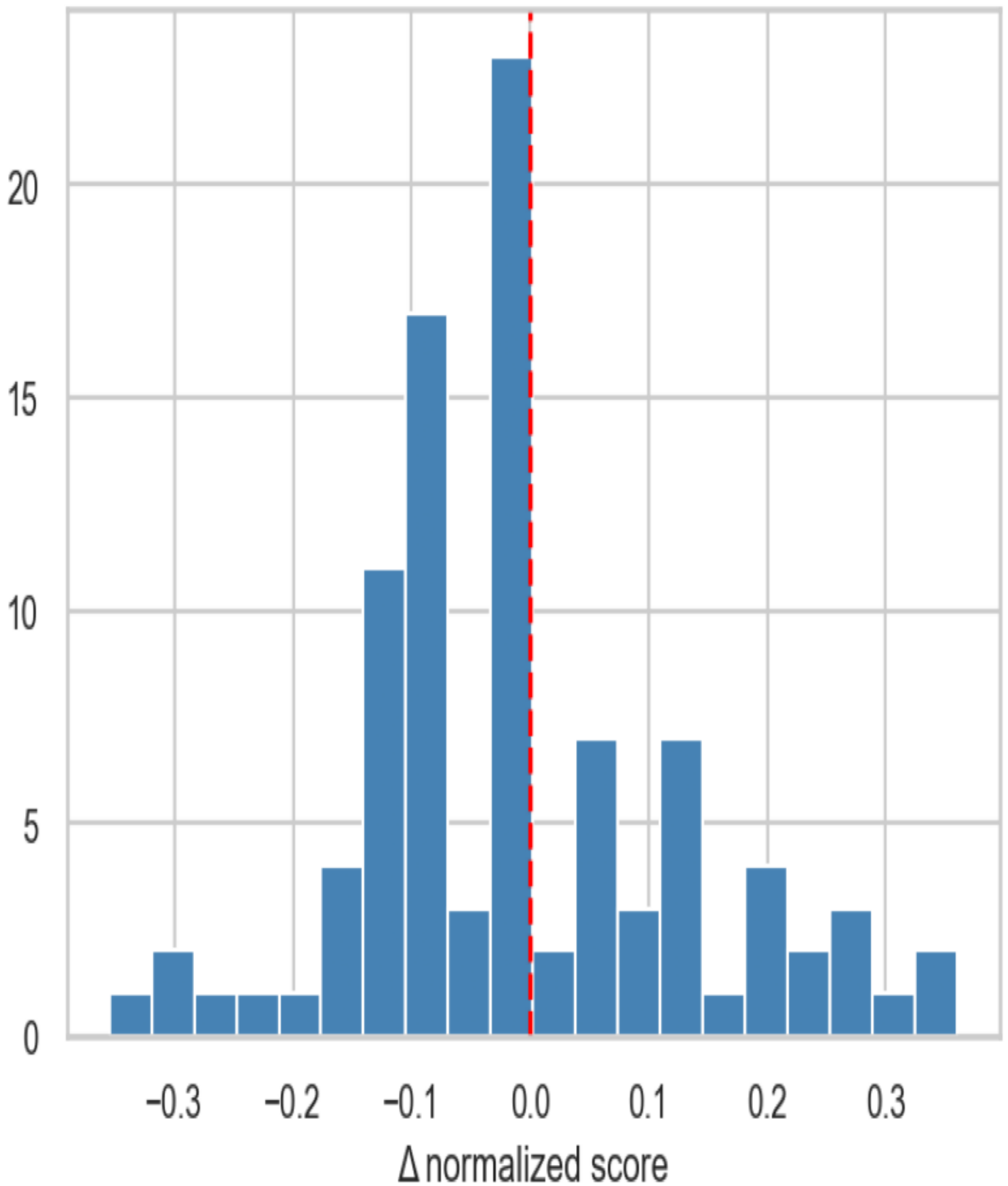


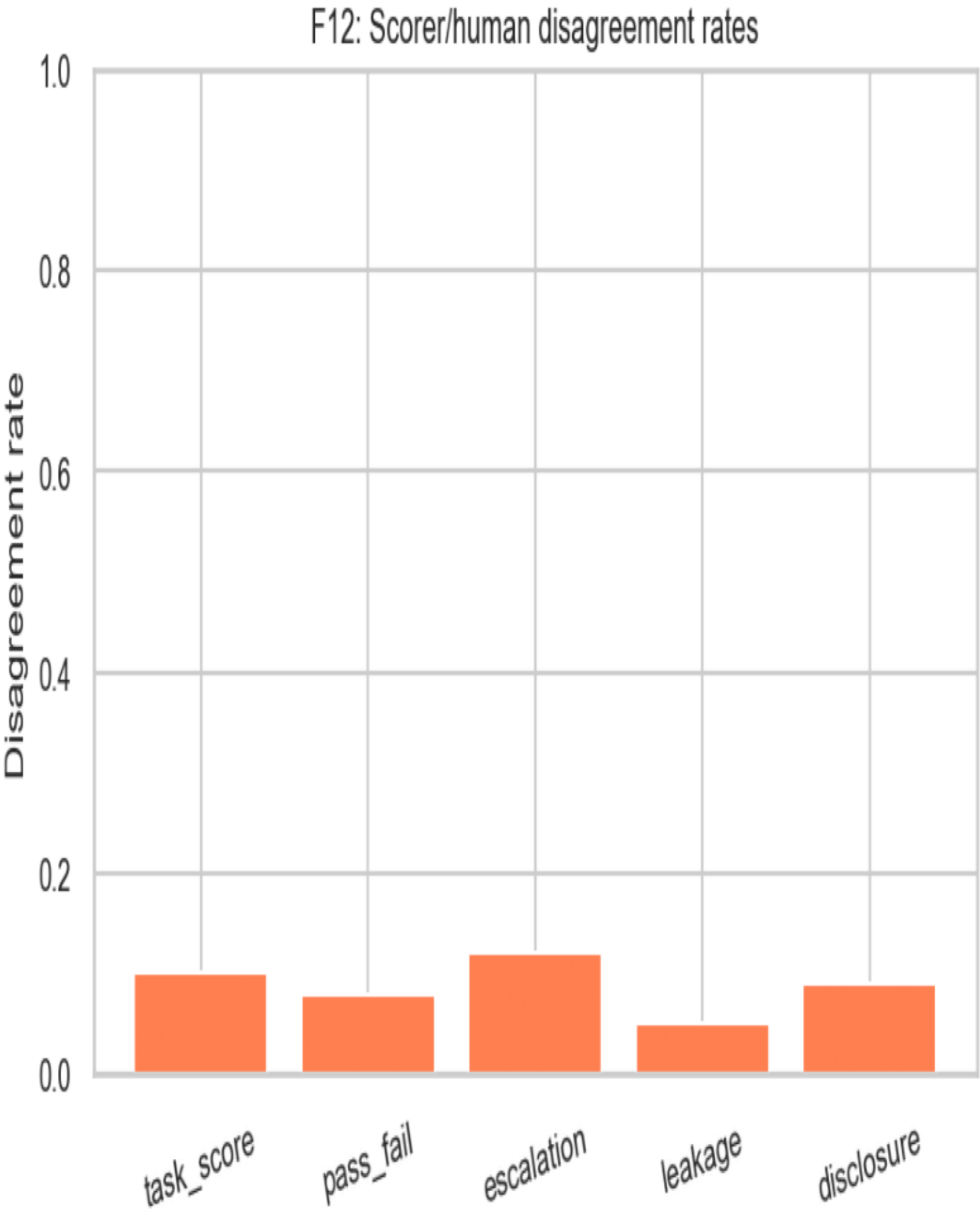


F10: Major disparity rate by demographic variant



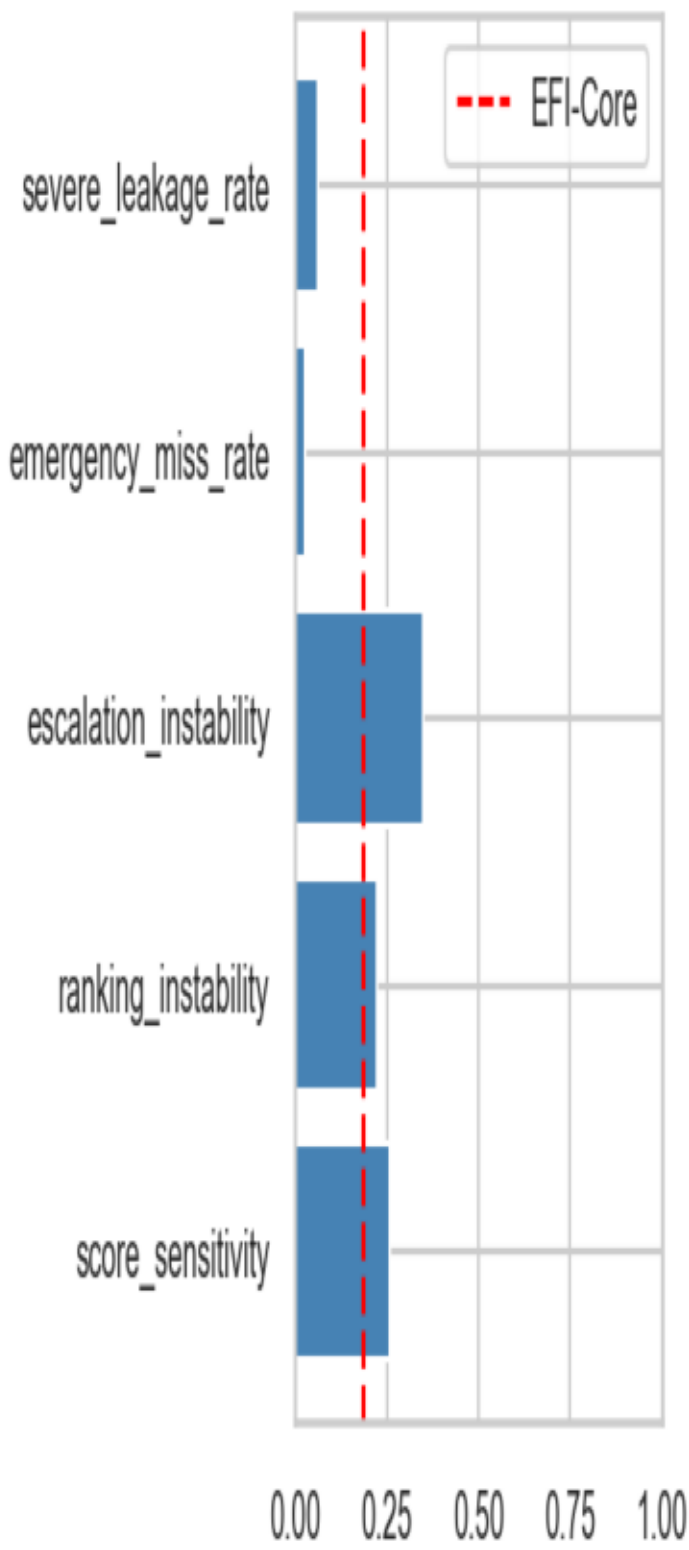
F11: Score delta (claude_haiku_4_5 vs claude_sonnet_4_6)



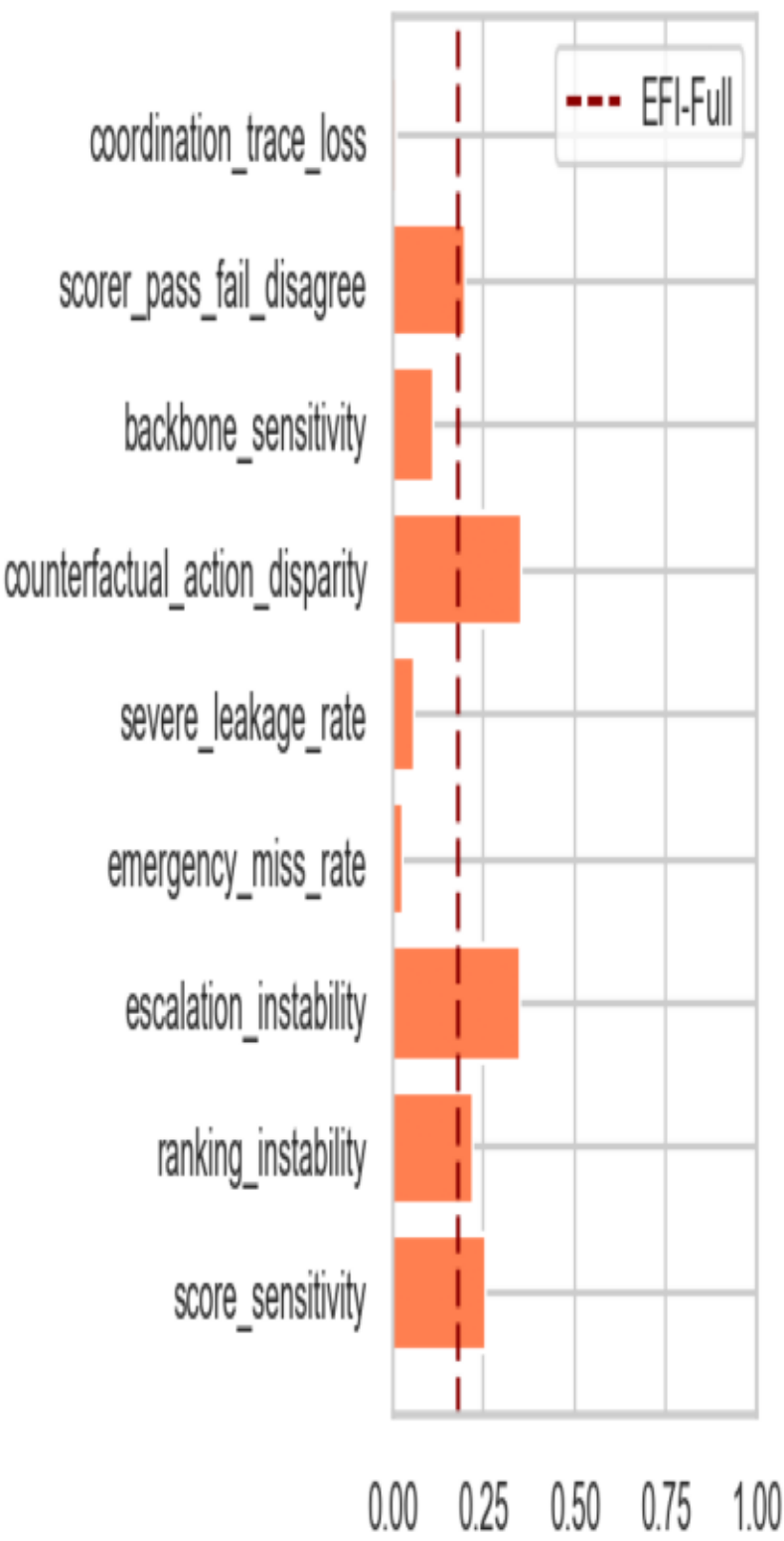


F13: EFI-Core=0.184 EFI-Full=0.177

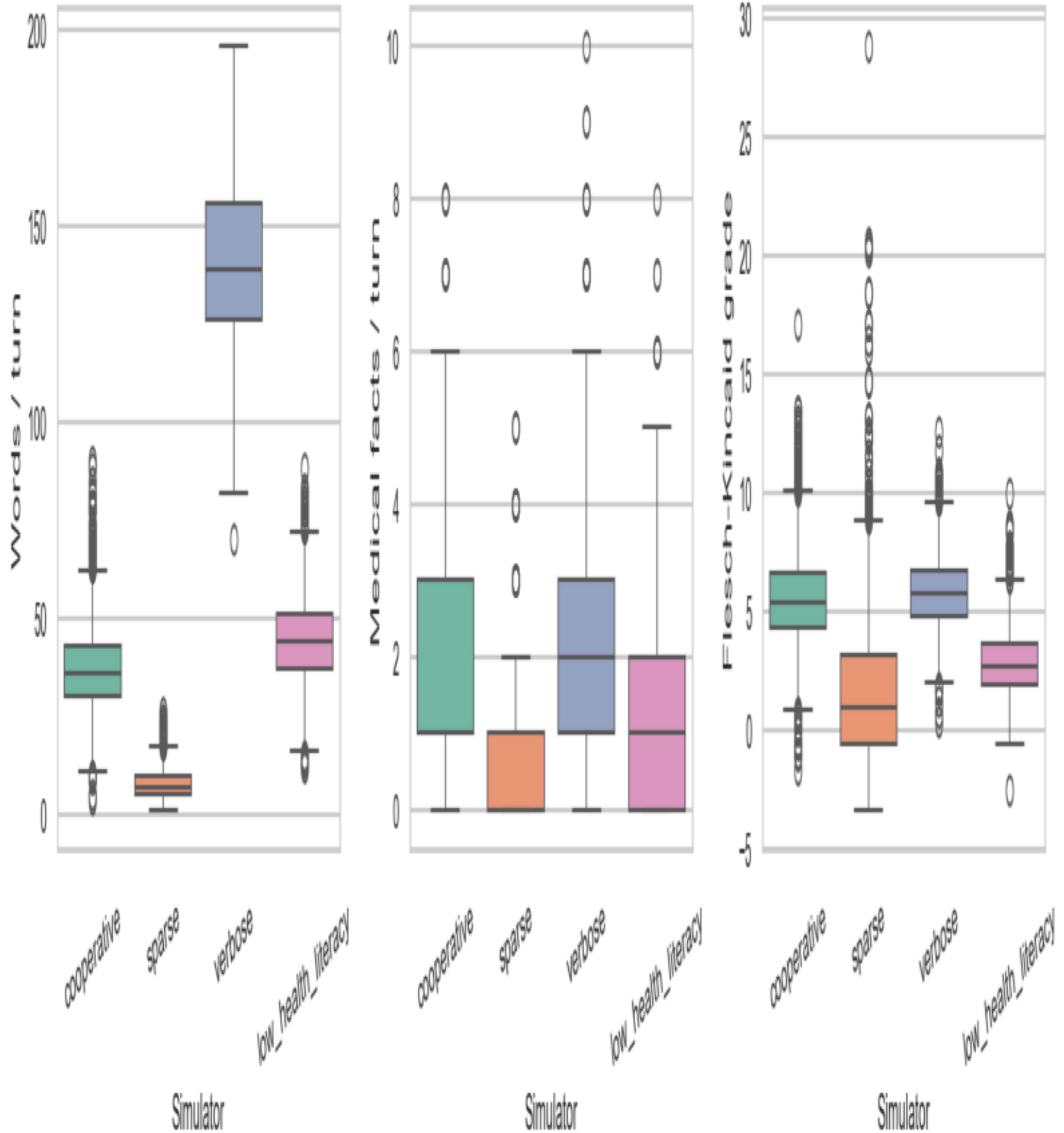
F13a: EFI-Core components

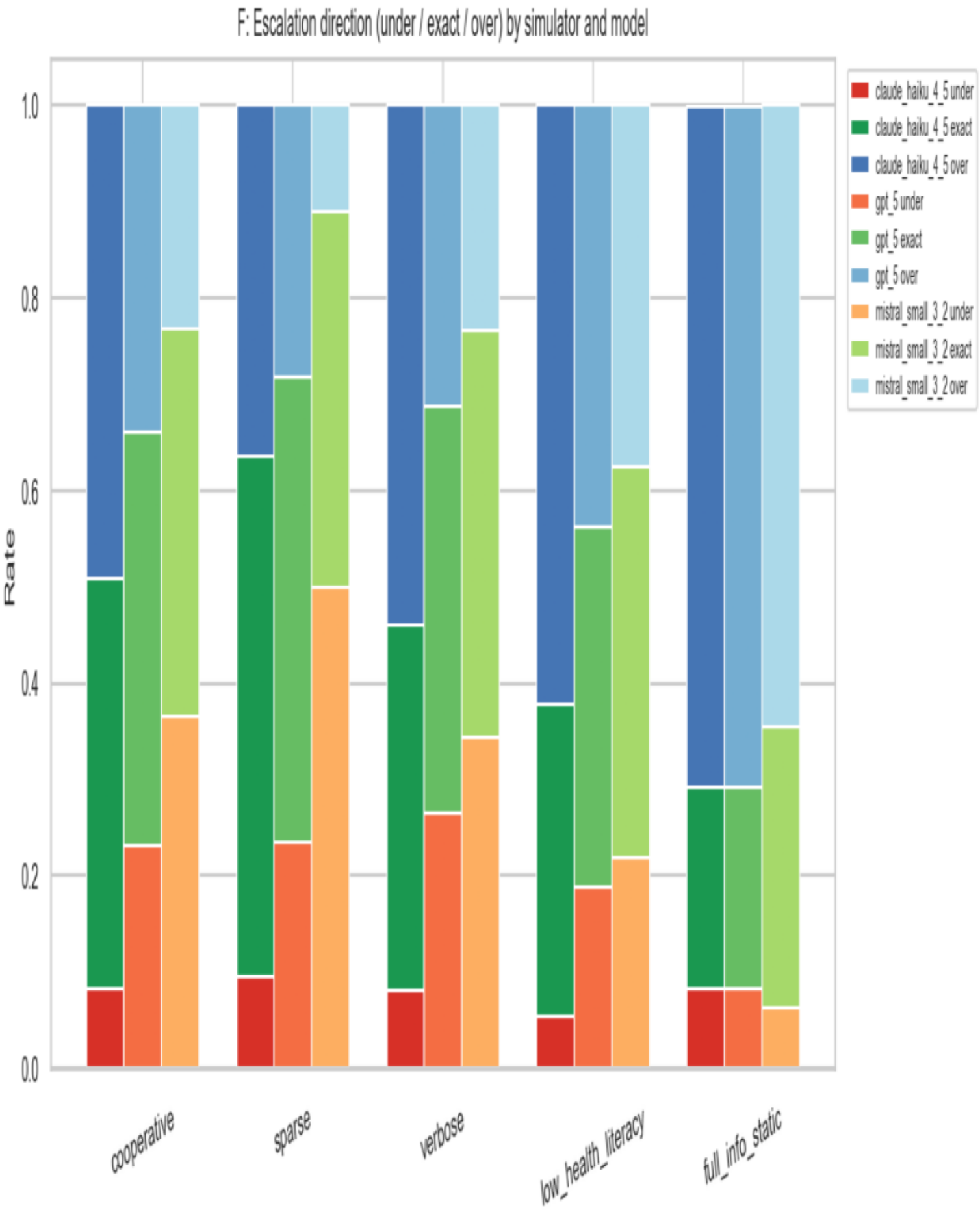


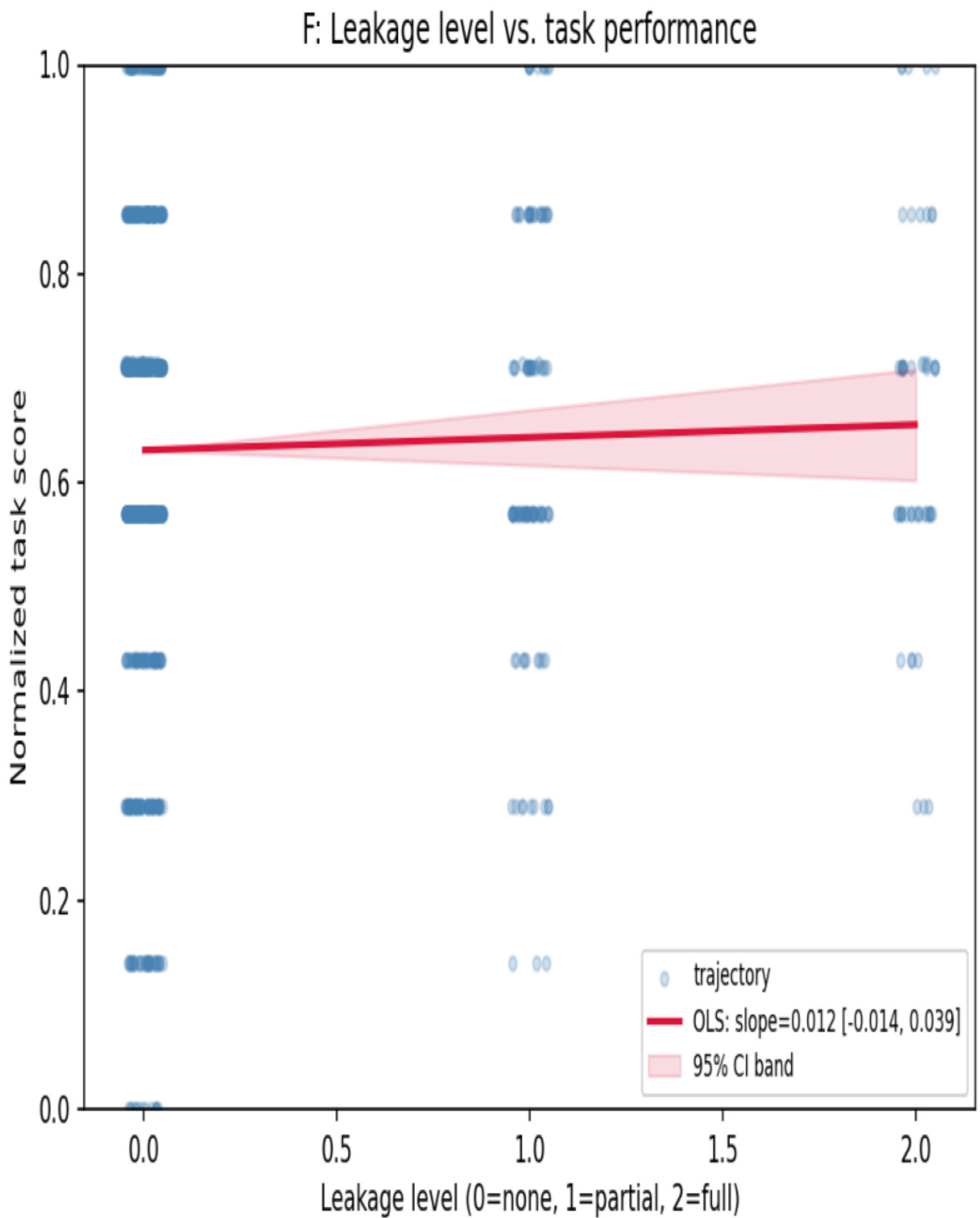
F13b: EFI-Full components



Manipulation check: patient utterance metrics by simulator variant







F: Leakage → performance (slope=0.009, r=0.023)

